A · barrel vs endcap · round 1

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barrel vs endcap · 98 features

  separates in barrel only    16
  separates in both           40
  separates in endcap only    18
  separates in neither        24

pages: this cover, paired ranking chart, cross-region table (top 60 by |delta|),
mk4 list features across regions.
per feature: features/NNN_<feature>/ (NNN = rank by |delta|) -> shape page + separation page
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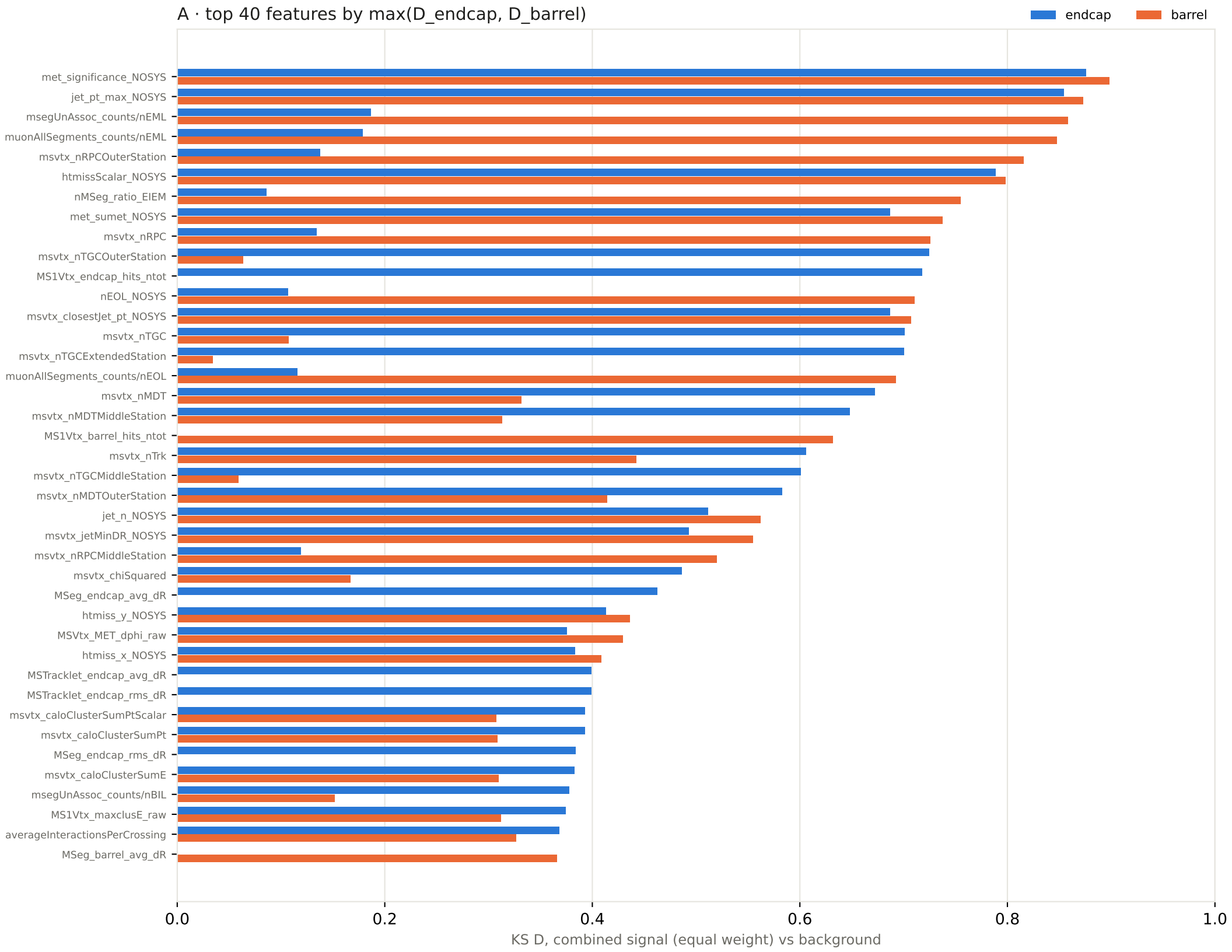
A · top 40 features by max(D_endcap, D_barrel)
endcap
barrel
met_significance_NOSYS
jet_pt_max_NOSYS
msegUnAssoc_counts/nEML
muonAllSegments_counts/nEML
msvtx_nRPCOuterStation
htmissScalar_NOSYS
nMSeg_ratio_EIEM
met_sumet_NOSYS
msvtx_nRPC
msvtx_nTGCOuterStation
MS1Vtx_endcap_hits_ntot
nEOL_NOSYS
msvtx_closestJet_pt_NOSYS
msvtx_nTGC
msvtx_nTGCExtendedStation
muonAllSegments_counts/nEOL
msvtx_nMDT
msvtx_nMDTMiddleStation
MS1Vtx_barrel_hits_ntot
msvtx_nTrk
msvtx_nTGCMiddleStation
msvtx_nMDTOuterStation
jet_n_NOSYS
msvtx_jetMinDR_NOSYS
msvtx_nRPCMiddleStation
msvtx_chiSquared
MSeg_endcap_avg_dR
htmiss_y_NOSYS
MSVtx_MET_dphi_raw
htmiss_x_NOSYS
MSTracklet_endcap_avg_dR
MSTracklet_endcap_rms_dR
msvtx_caloClusterSumPtScalar
msvtx_caloClusterSumPt
MSeg_endcap_rms_dR
msvtx_caloClusterSumE
msegUnAssoc_counts/nBIL
MS1Vtx_maxclusE_raw
averageInteractionsPerCrossing
MSeg_barrel_avg_dR
0.0
0.2
0.4
0.6
0.8
1.0
KS D, combined signal (equal weight) vs background

A · cross-region table (sorted by |delta|) (1/2)

feature	D_end	D_bar	delta	rk_e	rk_b	shapeB	shapeS	separates
msvtx_nRPCOuterStation	0.138	0.815	+0.677	66	5	0.998	0.985	barrel only
msegUnAssoc_counts/nEML	0.187	0.858	+0.672	61	3	0.935	0.976	barrel only
nMSeg_ratio_EIEM	0.086	0.755	+0.669	81	7	0.746	0.842	barrel only
muonAllSegments_counts/nEML	0.179	0.848	+0.669	62	4	0.937	0.975	barrel only
msvtx_nTGCExtendedStation	0.700	0.034	-0.666	7	88	1.000	1.000	endcap only
msvtx_nTGCOuterStation	0.725	0.063	-0.661	4	84	1.000	1.000	endcap only
nEOL_NOSYS	0.107	0.710	+0.604	76	10	0.994	0.990	barrel only
msvtx_nTGC	0.701	0.108	-0.593	6	75	1.000	1.000	endcap only
msvtx_nRPC	0.134	0.725	+0.591	68	9	1.000	0.990	barrel only
muonAllSegments_counts/nEOL	0.116	0.692	+0.576	72	12	0.991	0.986	barrel only
msvtx_nTGCMiddleStation	0.601	0.059	-0.542	13	85	1.000	1.000	endcap only
msvtx_nRPCMiddleStation	0.119	0.520	+0.401	69	16	0.998	0.980	barrel only
msvtx_nMDT	0.672	0.331	-0.341	10	25	0.246	0.302	both
msvtx_nMDTMiddleStation	0.648	0.313	-0.335	11	30	0.538	0.495	both
msvtx_chiSquared	0.486	0.167	-0.319	17	60	0.552	0.453	endcap only
muonAllSegments_counts/nBML	0.062	0.292	+0.229	83	39	0.944	0.907	barrel only
msegUnAssoc_counts/nBIL	0.378	0.152	-0.226	27	65	0.444	0.535	endcap only
muonAllSegments_counts/nBIL	0.357	0.146	-0.211	36	66	0.512	0.547	endcap only
msegUnAssoc_counts/nBML	0.069	0.277	+0.207	82	44	0.932	0.903	barrel only
msvtx_phi_NOSYS	0.099	0.285	+0.186	78	40	0.345	0.049	barrel only
msvtx_nMDTOuterStation	0.583	0.414	-0.169	14	20	0.380	0.298	both
msvtx_nTrk	0.606	0.442	-0.163	12	17	0.076	0.225	both
msvtx_nMDTInnerStation	0.312	0.154	-0.157	44	64	0.906	0.770	endcap only
msvtx_nMDTInwards	0.269	0.114	-0.155	50	72	0.189	0.190	endcap only
msegUnAssoc_nSeg	0.187	0.314	+0.127	60	29	0.297	0.414	barrel only
msvtx_muRoIClusTrig_mindR_NOSYS	0.232	0.119	-0.114	54	71	0.055	0.142	endcap only
msvtx_el4ecal	0.337	0.243	-0.094	39	52	0.292	0.102	both
msvtx_el1ecal	0.324	0.235	-0.089	42	55	0.161	0.077	both
msvtx_caloClusterSumPtScalar	0.393	0.307	-0.085	22	34	0.110	0.031	both
msvtx_nCaloClusters_Egt300MeV_NOSYS	0.366	0.281	-0.085	31	43	0.109	0.026	both
msvtx_caloClusterSumPt	0.393	0.308	-0.084	23	33	0.111	0.031	both
msvtx_nCaloClusters_NOSYS	0.365	0.281	-0.083	33	42	0.107	0.024	both
MS1Vtx_maxclustime_raw	0.356	0.274	-0.082	37	45	0.101	0.037	both
msvtx_nRPCInwards	0.105	0.024	-0.082	77	90	0.207	0.120	neither
mseg_counts/nBML	0.053	0.135	+0.082	86	68	0.596	0.594	neither
muonAllSegments_counts/nBOL	0.253	0.333	+0.080	51	24	1.000	0.950	both
msvtx_trackHardMinDR_NOSYS	0.243	0.320	+0.077	53	27	0.066	0.148	both
mseg_counts/nEIL	0.093	0.170	+0.077	80	58	0.262	0.305	neither
msvtx_el2ecal	0.343	0.267	-0.076	38	47	0.221	0.087	both
mseg_counts/nBOL	0.057	0.132	+0.075	85	69	0.647	0.626	neither
MS1Vtx_rms_clustime_raw	0.335	0.262	-0.074	40	50	0.096	0.043	both
msvtx_caloClusterSumE	0.383	0.310	-0.073	26	32	0.385	0.118	both
muonSegments_nSeg	0.201	0.274	+0.073	58	46	0.285	0.401	both
MS1Vtx_avg_clustime_raw	0.229	0.157	-0.072	55	63	0.064	0.031	endcap only
msvtx_el4hcal	0.094	0.166	+0.072	79	61	0.099	0.032	neither
msvtx_el3ecal	0.364	0.293	-0.070	34	38	0.268	0.096	both
nMSeg_ratio_BIBM	0.330	0.263	-0.068	41	49	0.553	0.397	both
MS1Vtx_rms_clusE_raw	0.365	0.299	-0.066	32	36	0.434	0.197	both
MS1Vtx_maxclusE_raw	0.374	0.312	-0.062	29	31	0.354	0.152	both
msvtx_jetMinDR_NOSYS	0.493	0.555	+0.062	16	15	0.041	0.073	both [htmiss-linked]
actualInteractionsPerCrossing	0.298	0.237	-0.061	47	54	0.075	0.015	both [non-physics]
MS1Vtx_avg_clusE_raw	0.361	0.304	-0.057	35	35	0.485	0.215	both
mseg_counts/nEML	0.033	0.090	+0.057	91	81	0.425	0.437	neither
MS1Vtx_mindR_jetcut_raw	0.308	0.365	+0.057	46	23	0.027	0.076	both
mseg_counts/nBIL	0.041	0.097	+0.056	88	79	0.521	0.378	neither
MSVtx_MET_dphi_raw	0.375	0.429	+0.054	28	19	0.026	0.055	both
msvtx_el2hcal	0.311	0.259	-0.052	45	51	0.119	0.037	both
track_scalar_sum_pt_barrel	0.213	0.265	+0.052	57	48	0.058	0.040	both
jet_n_NOSYS	0.511	0.562	+0.051	15	14	0.009	0.064	both [htmiss-linked]
met_sumet_NOSYS	0.687	0.737	+0.050	9	8	0.026	0.103	both [htmiss-linked]

A · cross-region table (sorted by |delta|) (2/2)

separates = D >= max(0.20, 2 x noise floor) in that region
shapeB / shapeS = KS D between barrel and endcap distributions of background / combined signal

A · mk4 list features in both regions (1/2)

NN1	barrel	nMSeg_ratio_BIBM	endcap	0.330	barrel	0.263	both
NN1	barrel	track_scalar_sum_pt_barrel	endcap	0.213	barrel	0.265	both
NN1	barrel	MS1Vtx_clusE	endcap	0.383	barrel	0.310	both
NN1	barrel	MS1Vtx_avg_clusE	endcap	0.361	barrel	0.304	both
NN1	barrel	MS1Vtx_rms_clusE	endcap	0.365	barrel	0.299	both
NN1	barrel	MS1Vtx_maxclusE	endcap	0.374	barrel	0.312	both
NN1	barrel	MS1Vtx_l1ecal	endcap	0.324	barrel	0.235	both
NN1	barrel	MS1Vtx_l1hcal	endcap	0.313	barrel	nan	endcap only
NN1	barrel	MS1Vtx_l2ecal	endcap	0.343	barrel	0.267	both
NN1	barrel	MS1Vtx_l2hcal	endcap	0.311	barrel	0.259	both
NN1	barrel	MS1Vtx_l3ecal	endcap	0.364	barrel	0.293	both
NN1	barrel	MS1Vtx_l3hcal	endcap	0.193	barrel	0.240	barrel only
NN1	barrel	MS1Vtx_l4ecal	endcap	0.337	barrel	0.243	both
NN1	barrel	MS1Vtx_l4hcal	endcap	0.094	barrel	0.166	neither
NN1	barrel	MS1Vtx_mindR_jetcut	endcap	0.308	barrel	0.365	both
NN1	barrel	MS1Vtx_sumTrackPt0p2Cone	endcap	0.140	barrel	0.111	neither
NN1	barrel	msvtx_caloClusterSumPtScalar	endcap	0.393	barrel	0.307	both
NN1	barrel	msegUnAssoc_counts/nBIL	endcap	0.378	barrel	0.152	endcap only
NN1	barrel	msegUnAssoc_counts/nBML	endcap	0.069	barrel	0.277	barrel only
NN1	barrel	msegUnAssoc_counts/nBOL	endcap	0.271	barrel	0.319	both
NN1	endcap	nMSeg_ratio_EIEM	endcap	0.086	barrel	0.755	barrel only
NN1	endcap	track_scalar_sum_pt_endcap	endcap	0.286	barrel	0.284	both
NN1	endcap	MS1Vtx_clusE	endcap	0.383	barrel	0.310	both
NN1	endcap	MS1Vtx_avg_clusE	endcap	0.361	barrel	0.304	both
NN1	endcap	MS1Vtx_rms_clusE	endcap	0.365	barrel	0.299	both
NN1	endcap	MS1Vtx_maxclusE	endcap	0.374	barrel	0.312	both
NN1	endcap	MS1Vtx_l1ecal	endcap	0.324	barrel	0.235	both
NN1	endcap	MS1Vtx_l1hcal	endcap	0.313	barrel	nan	endcap only
NN1	endcap	MS1Vtx_l2ecal	endcap	0.343	barrel	0.267	both
NN1	endcap	MS1Vtx_l2hcal	endcap	0.311	barrel	0.259	both
NN1	endcap	MS1Vtx_l3ecal	endcap	0.364	barrel	0.293	both
NN1	endcap	MS1Vtx_l3hcal	endcap	0.193	barrel	0.240	barrel only
NN1	endcap	MS1Vtx_l4ecal	endcap	0.337	barrel	0.243	both
NN1	endcap	MS1Vtx_l4hcal	endcap	0.094	barrel	0.166	neither
NN1	endcap	MS1Vtx_mindR_jetcut	endcap	0.308	barrel	0.365	both
NN1	endcap	MS1Vtx_sumTrackPt0p2Cone	endcap	0.140	barrel	0.111	neither
NN2	barrel	MSTracklet_barrel_avg_dR	endcap	nan	barrel	0.097	neither
NN2	barrel	MSTracklet_barrel_rms_dR	endcap	nan	barrel	0.097	neither
NN2	barrel	MSeg_barrel_avg_dR	endcap	nan	barrel	0.366	barrel only
NN2	barrel	MSeg_barrel_rms_dR	endcap	nan	barrel	0.297	barrel only
NN2	barrel	MS1Vtx_barrel_hits_ntot	endcap	nan	barrel	0.632	barrel only
NN2	barrel	MSVtx_MET_dphi	endcap	0.375	barrel	0.429	both
NN2	barrel	MS1Vtx_maxclustime	endcap	0.356	barrel	0.274	both
NN2	barrel	MS1Vtx_avg_clustime	endcap	0.229	barrel	0.157	endcap only
NN2	barrel	MS1Vtx_rms_clustime	endcap	0.335	barrel	0.262	both
NN2	barrel	MS1Vtx_nTracklet	endcap	0.606	barrel	0.442	both
NN2	endcap	MSTracklet_endcap_avg_dR	endcap	0.399	barrel	nan	endcap only
NN2	endcap	MSTracklet_endcap_rms_dR	endcap	0.399	barrel	nan	endcap only
NN2	endcap	MSeg_endcap_avg_dR	endcap	0.462	barrel	nan	endcap only
NN2	endcap	MSeg_endcap_rms_dR	endcap	0.384	barrel	nan	endcap only
NN2	endcap	MS1Vtx_endcap_hits_ntot	endcap	0.718	barrel	nan	endcap only
NN2	endcap	MSVtx_MET_dphi	endcap	0.375	barrel	0.429	both

A · mk4 list features in both regions (2/2)

```
NN2 endcap  MS1Vtx_maxclustime            endcap  0.356  barrel  0.274  both
NN2 endcap  MS1Vtx_avg_clustime           endcap  0.229  barrel  0.157  endcap only
NN2 endcap  MS1Vtx_rms_clustime           endcap  0.335  barrel  0.262  both
NN2 endcap  MS1Vtx_nTracklet              endcap  0.606  barrel  0.442  both
```